

04.05 Solving Radical Equations

Common problems and solution methods

Rule guide: recognize the task first

Recognize	Apply these rules
A variable is under one radical.	Isolate the radical, square both sides, solve, and check.
Two radicals remain.	Isolate one radical at a time and repeat the process.
A candidate makes a radicand negative or fails the original equation.	Reject it as invalid or extraneous.

Example 1: Solve a radical equation

Problem

$$\sqrt{x+1} = x-1$$

How to solve it

Restrict:

$$x-1 \geq 0 \Rightarrow x \geq 1$$

Square and check:

$$x+1 = (x-1)^2 \Rightarrow x=2; \quad \sqrt{3} \neq 1, \text{ so reject } x=2$$

Example 2: Solve with a coefficient

Problem

$$2\sqrt{x+1} = 6$$

How to solve it

Isolate:

$$\sqrt{x+1} = 3$$

Square:

$$x+1 = 9 \Rightarrow x=8$$

Example 3: Solve two radicals

Problem

$$\sqrt{x+5} - \sqrt{x} = 1$$

How to solve it

Isolate and square:

$$\sqrt{x+5} = 1 + \sqrt{x}$$

Solve and check:

$x = 4$; substitute in the original equation